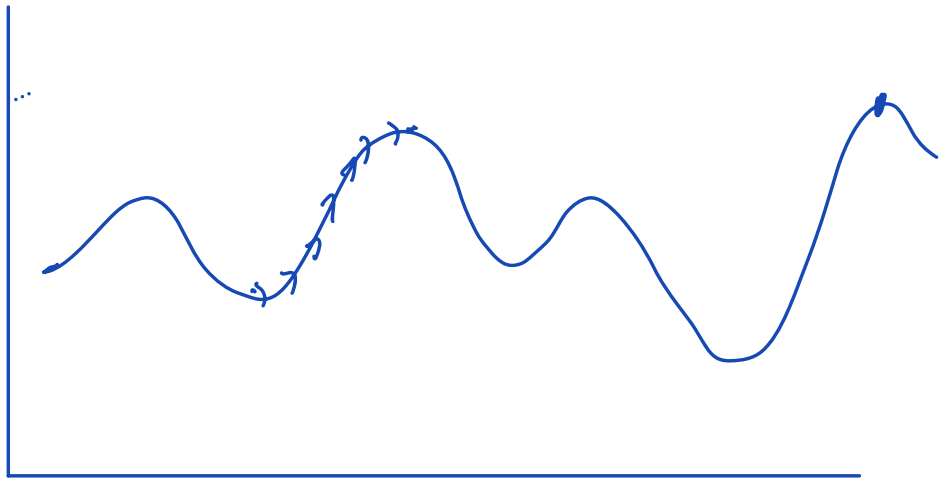
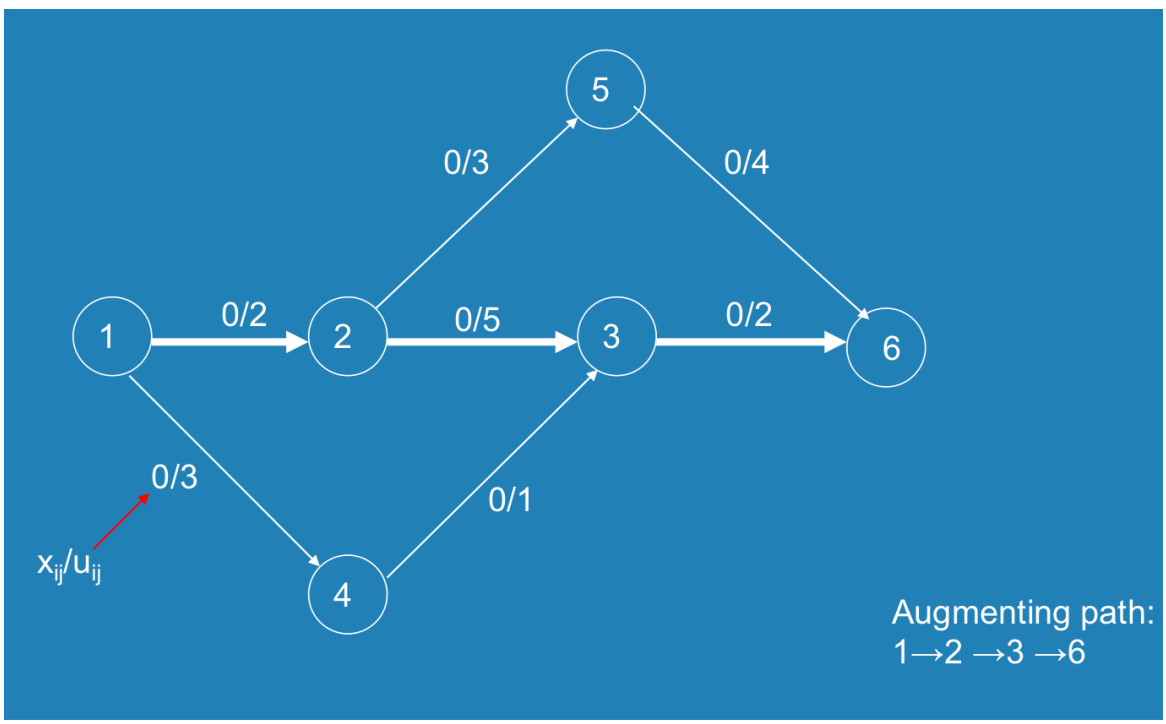
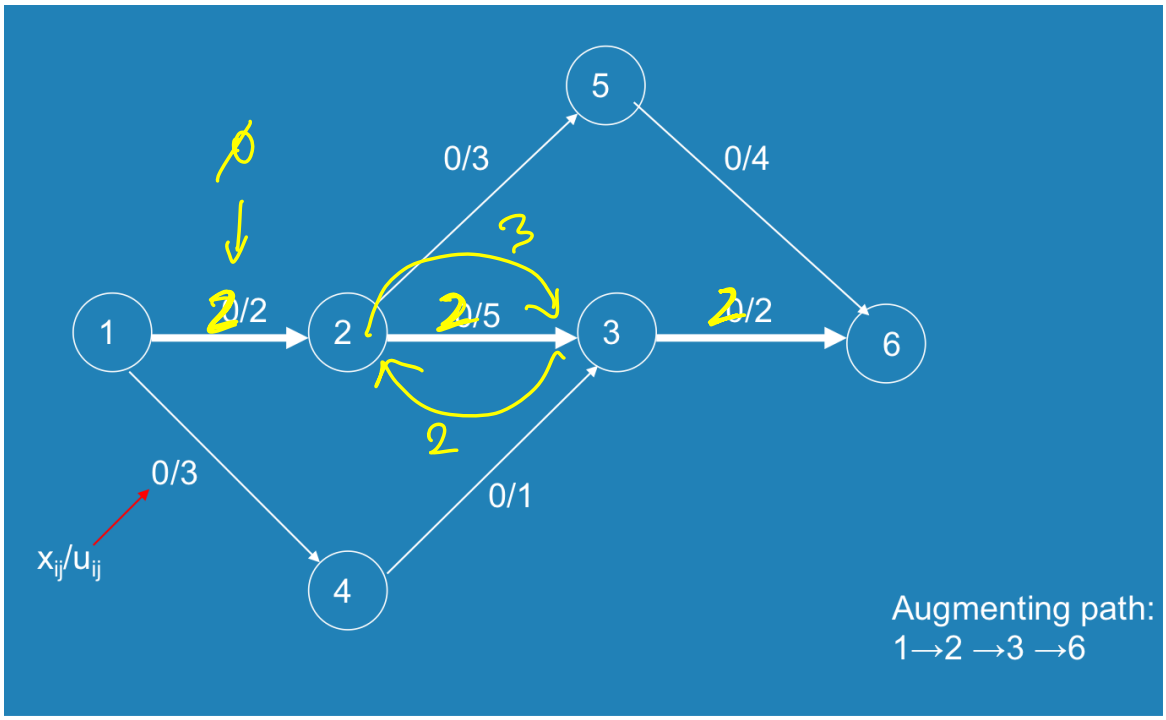
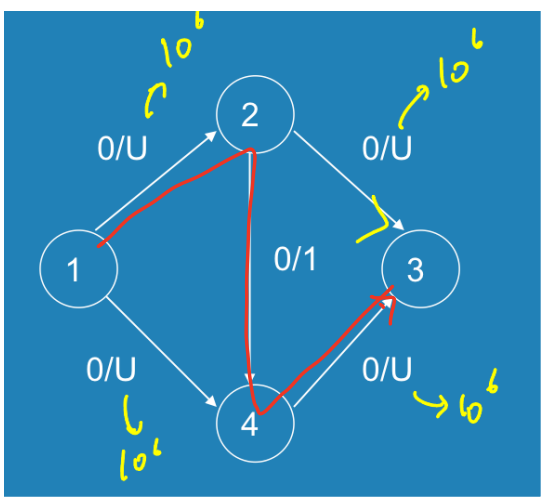


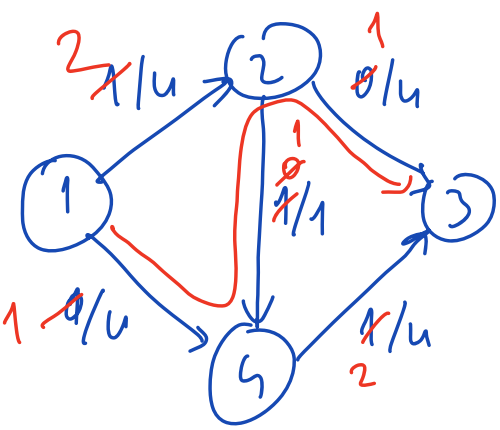

5
0/3
0/4
1
0/2
2
0/5
3
0/2
6
0/3
x_{ij}/u_{ij}
0/1
4
Augmenting path:
1→2 →3 →6





$1 \to 2 \to 4 \to 3$

$\min\{4, 1, 4\} = 1$



$1 \to 4 \leftarrow 2 \to 3$

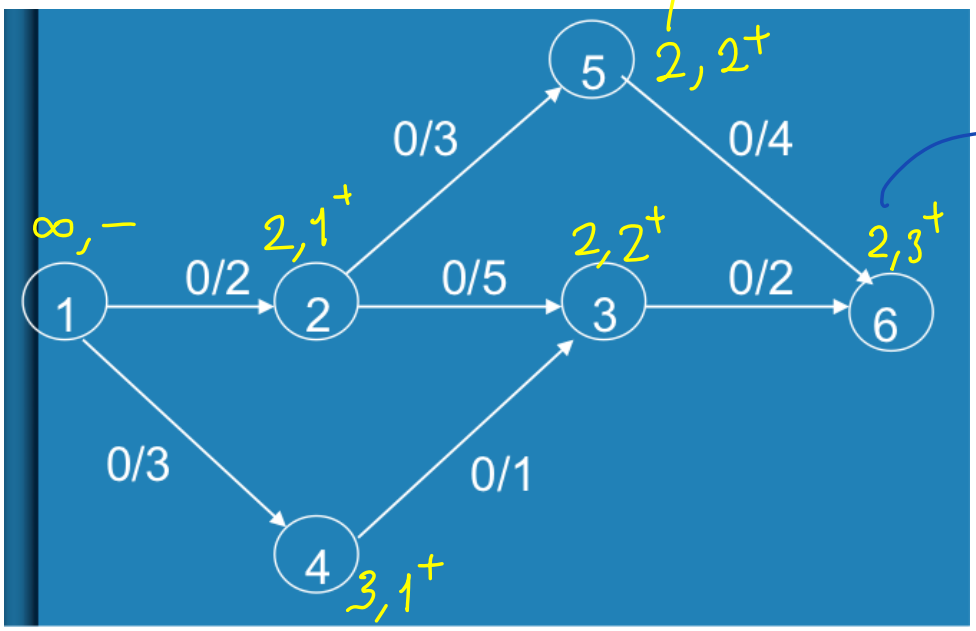
$\min\{4, 1, 4\} = 1$

$1 \to 2 \to 4 \to 3$ flow = 1

$1 \to 4 \leftarrow 2 \to 3$ flow = 1

⋮

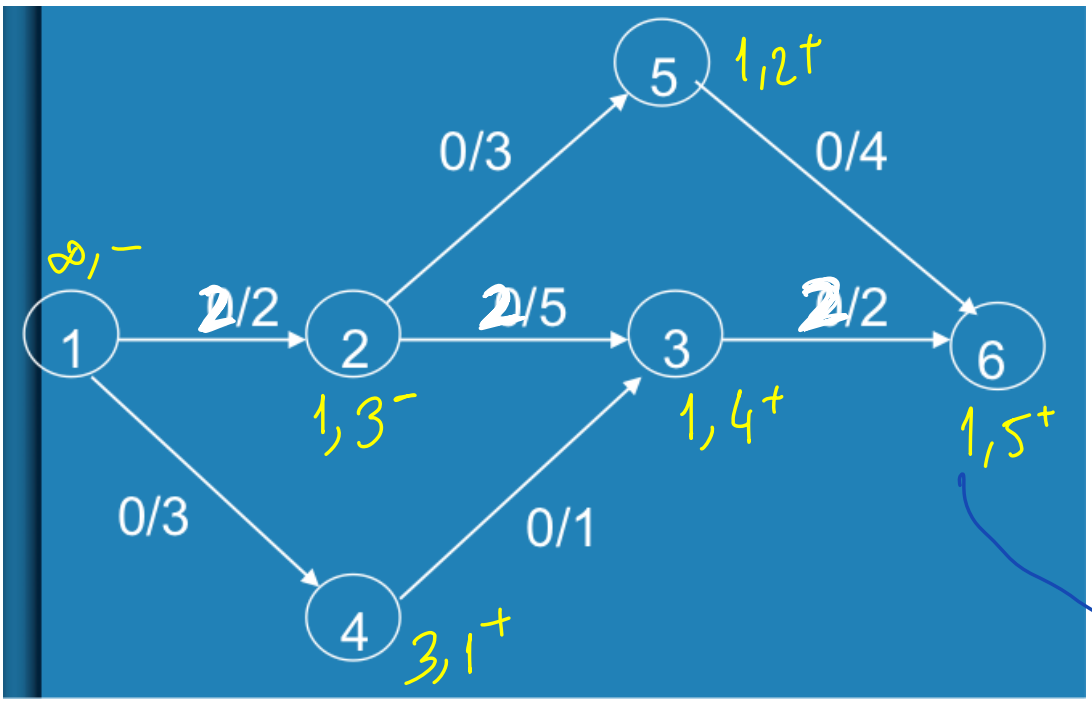
after 20 iterations, it converges to the maximum flow.



Q: 1 2 4 3 5 6

$1 \to 2 \to 3 \to 6$

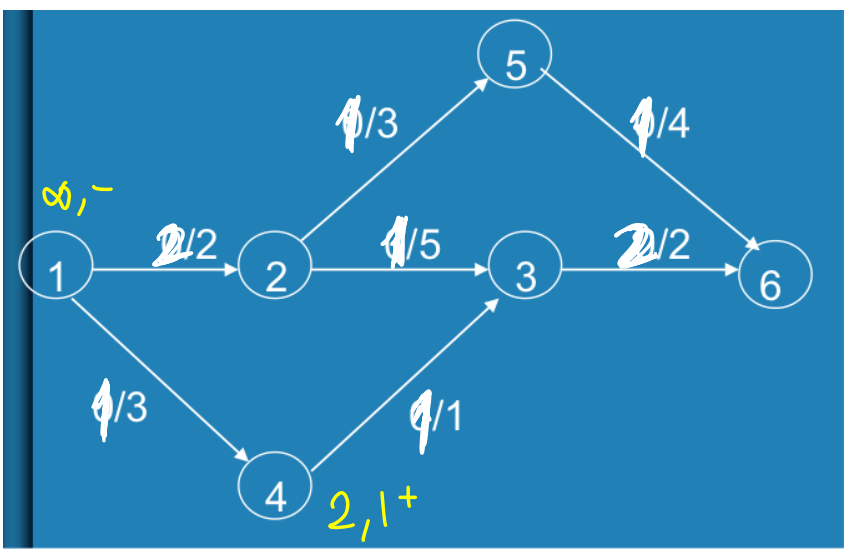
flow = 2



Q: 1 4 3 2 5 6

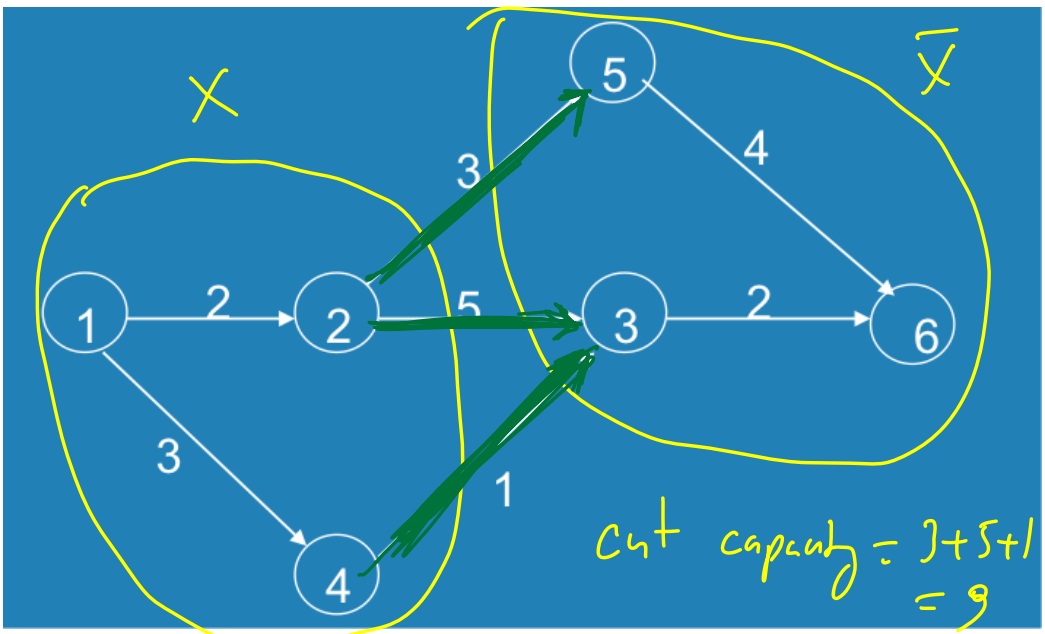
1 → 4 → 3 ← 2 → 5 → 6

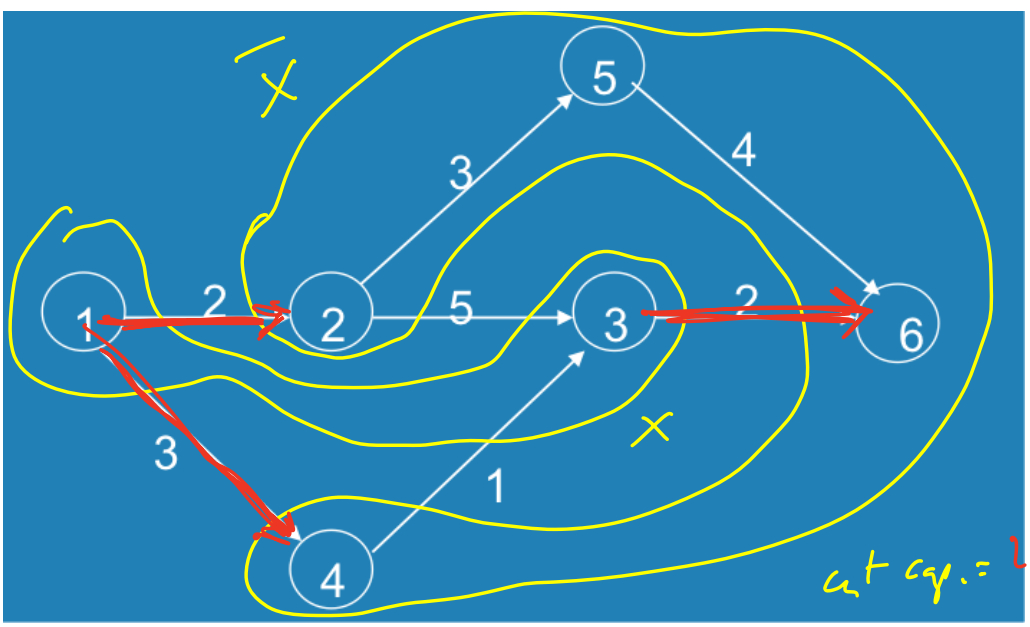
flow = 1



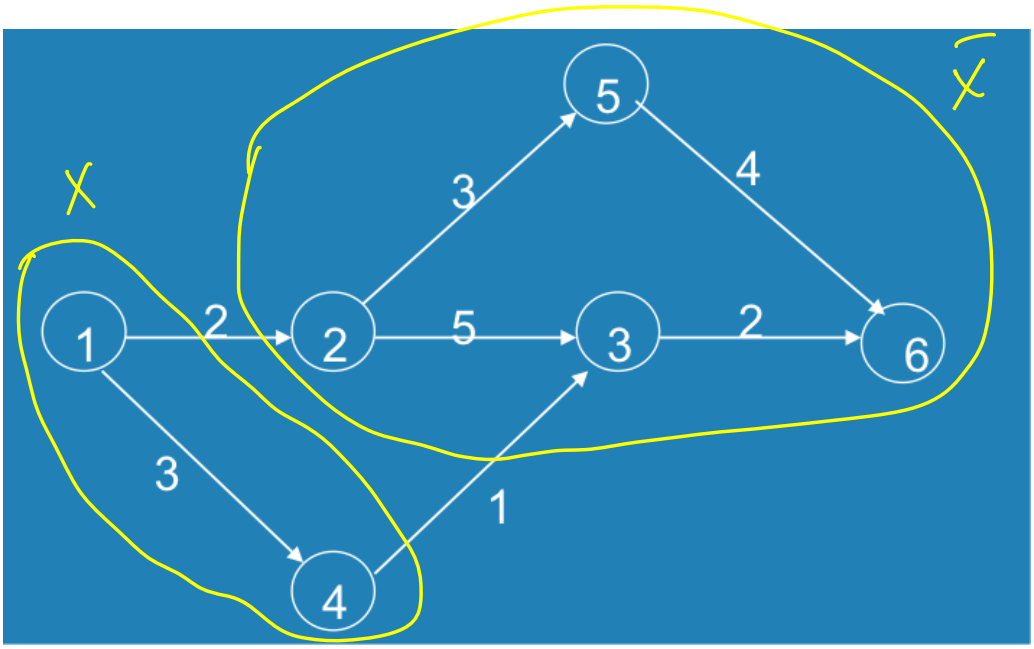
Q: 1 4

no aug. path. Stop!



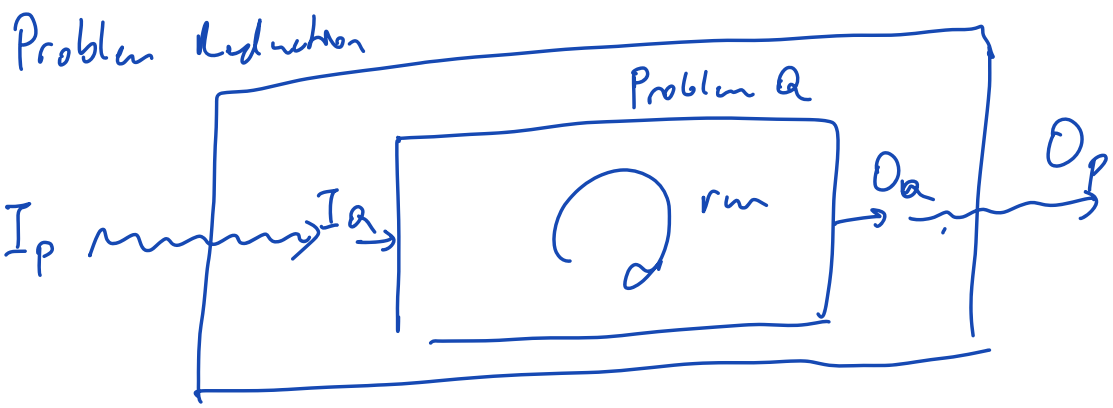


$X = \{1, 3\}$ $\bar{X} = \{2, 4, 5, 6\}$



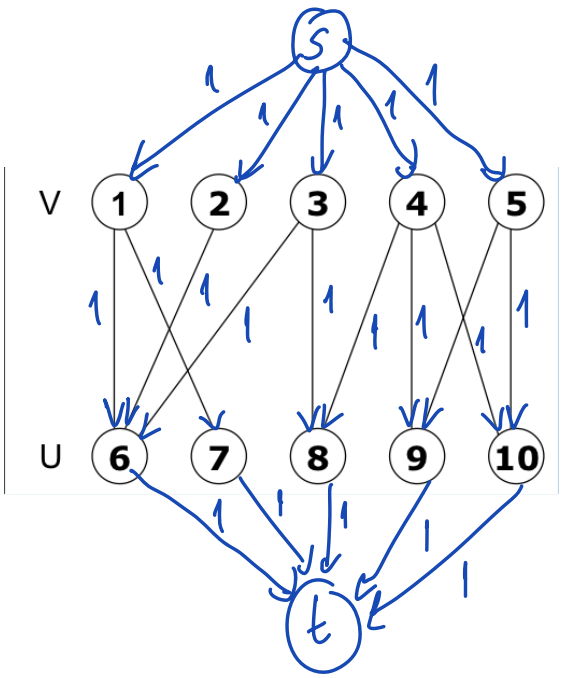
$X = \{1, 4\}$ $\bar{X} = \{2, 3, 5, 6\}$

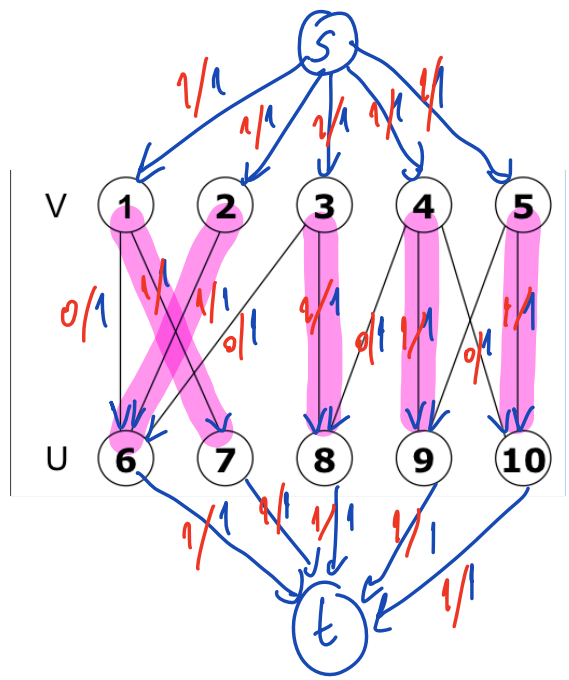
cut $= \{(1,2), (4,3)\}$ cut cap. $= 1+2 = 3$ //



Q: Max Flow Problem

p: max Matching Problem.





$s \to 1 \to 6 \to t$

$s \to 3 \to 8 \to t$

$s \to 4 \to 9 \to t$

$s \to 5 \to 10 \to t$

$s \to 2 \to 6 \leftarrow 1 \to 7 \to t$